Backend Engineering Standards and Code Review Policy

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Owning Department: Backend Engineering
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System of record: Confluence, Policies space, and ZenPeople (acknowledgement)
Related policies: ZEN-POL-006 (Information Security Policy); ZEN-DPL-DCL-01 (Cloud Operations, On-call and Incident Management Policy); ZEN-DPL-QAT-01 (Quality Gates and Release Sign-off Policy)

Policy History

Revision	Amendment description	Effective date	Reviewed by	Approved by
1	Engineering standards written	1 May 2019	Director of Engineering, Backend	Co-Founder & CTO
2.0	Rewritten after Project Atlas	1 Apr 2024	Principal Backend Engineer	Co-Founder & CTO
2.1	Dependency patching timelines tightened	6 Jan 2025	Principal Backend Engineer	Co-Founder & CTO

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1. Objective

To keep backend services reliable, secure and easy to change through consistent engineering practice.

2. Applies to

All engineers who change backend code or services.

3. Languages and stack

Approved languages are Go and Python (FastAPI). Any other language or datastore needs an Architecture Decision Record (ADR) approved by the Principal Backend Engineer and the Director of Engineering.

4. Branching and pull requests

- Trunk-based development with branches that live no longer than 2 days.
- Pull requests should be under 400 changed lines where possible.
- Every change needs CI to pass: build, tests, lint and security scan.
- Review rule: 1 approval for most services. 2 approvals, including one Principal Engineer or Director, for services in the billing, payments and authentication paths.

5. Quality bar

- New code must have at least 80 percent unit test coverage.
- Public APIs are versioned. Deprecated public API versions get at least 6 months' notice.
- Database migrations must be backward compatible (expand then contract).
- Secrets live in AWS Secrets Manager, never in code.

6. Service-level objectives

Every service defines SLOs. Defaults are 99.9 percent monthly availability and p95 latency under 250 ms. Services must expose metrics, logs and traces.

7. Dependencies

Dependency updates are reviewed monthly. Critical vulnerabilities are patched within 7 days and high ones within 14 days.

8. Ownership

Each service has a CODEOWNERS entry and a named on-call owner. Service owners join the on-call rota described in ZEN-DPL-DCL-01.

9. Disclaimer

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